can operate a cobot; in fact, certain end effectors allow you to swap between tools several times a day."

Today, cobots can be easily programmed to perform one or more tasks using user-friendly software, mobile apps, and even through physical movement or "lead-through programming" as ABB calls it. You can just hold the robotic arm and move it physically to 'train' it to do something—just like you would teach a kid yoga! It can also be reprogrammed as easily as that—so the same cobot can do multiple tasks as and when required.

Social robot Pepper helping with group therapy at Changi General Hospital, Singapore (Courtesy: Changi General Hospital)

There are various tools available—grippers such as finger-grippers for pick-and-place operations, end-of-arm tooling like glue dispensers, automatic tool changers, range extenders, supply systems, vision and software tools—which can be used to adapt the cobot for doing multiple chores.

"The best part about a cobot is that it can learn from the master and do the taught task even better than the master," quips Gurappa.

Technologies for better collaboration

Apart from mechanical and material improvements, advancements in technologies like AI, machine learning (ML), sensors, data analytics, and open source software are having a significant impact on the emerging generation of cobots. The experts we interviewed spoke of some of the significant technologies influencing cobots today.

AI and ML. "Advancements in ML are fuelling the growing deployment of cobots. Applications being rolled out allow cobots to mimic human behaviour—much as any human trainee would. This saves the time that previously would have been required for programming," says Sougandh.

Guruprasad adds, "Technologies like big data and AI are at play to develop bots such that one day they may be able to perform unrestricted movements they've figured out on their own. Apart from that, industrial cobots, sometimes, must work in harsh conditions that affect their data connection, so durability is quite necessary on that front. Heavy duty conductors (HDCs) are designed for this, and we're working on further innovations, since reliable connectivity is quite necessary. With their always-on connection to big data and AI, collaborative robots may eventually be able to perform unrestricted movements they've figured out on their own. This breakthrough will further boost their viability in less-structured manufacturing environments."

Cameras and sensors. Advanced sensors and cameras are essential to enable precision, learning, performance, and safety. Scott says regarding the cameras, "The rise of robust, affordable, smart cameras has been critical. The past few years have seen a number of camera systems come to market that feature depth sensors and integrated deep learning co-processors. These cameras make building a cobot cost-effective and improve its overall reliability."

Singrow, a Singaporean agri-tech company that harnesses technologies like vertical farming and robotics to produce healthy fruits and vegetables without pesticides, partnered with Universal Robots and Augmentus, a leader in easy-to-use and rapid AI-robotics software, to automate pollination and harvesting in their strawberry farm. "The company deployed a UR cobot adapted with a camera that would identify flowers using Augmentus' integrated AI technology for effective pollination and the picking of ripe strawberries," recalls Sougandh.

Open source software. "Open source software is one of the most important developments in the rise of collaborative robots. When the robotics industry was largely focused on industrial robots, each robot had its own software and hardware. Due to the proprietary nature of this software, there was little or no collaboration among industrial robots, and for decades not much innovation. In the same way that open-source initiatives like Linux, Apache and others gave rise to the Web, software initiatives such as the Robot Operating System have given rise to collaborative robots and more advanced robotics applications," says Scott.

ROS 2 is the second generation of ROS, and it has been built from the ground up to work in industrial settings. As ROS 2 becomes more prevalent, more and more vendors have begun to support it, thus simplifying system integration. micro-ROS is a variant of ROS built for embedded microcontrollers, which make it even easier to build robot components that support ROS out of the box.

Miniaturisation. People want more compact designs that are even stronger—and collaborative robots are no exception to that. Guruprasad says, "Miniaturisation of cobots also opens unique opportunities for electro-mechanical helpers that can exist outside huge production facilities. Institutions and businesses could benefit from miniaturised cobots with the intelligence to complete day-to-day mundane tasks."

"Technologies that would contribute to higher levels of collaboration